Daniel Sabey

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EDUCATION

PhD Candidate of Economics Cornell University *August 2021 - Present*

Chair: Mike Lovenheim

Committee Members: Sule Alan and Evan Riehl

MA Economics Cornell University *August 2023*

BS Economics Brigham Young University (3.97 GPA) *April 2020*

Minors in Math, Business, and Korean (4.0 GPA)

Brigham Young University Academic Full Tuition Scholarship for 4 Years

Magna Cum Laude

PUBLISHED PAPERS

"Decreasing time to baccalaureate degree in the United States." With Denning, J. T., Eide, E. R., & Mumford, K. J., (2022). *Economics of Education Review, 90,* 102287.

After increasing in the 1970s and 1980s, time to bachelor's degree has declined since the 1990s. We document this fact using data from three nationally representative surveys. We show that this pattern is occurring across school types and for all student types. Using administrative student records from 10 large universities, we confirm the finding and show that it is robust to alternative sample definitions. We discuss what might explain the decline in time to bachelor's degree by considering trends in student preparation, state funding, student enrollment, study time, and student employment during college.

WORKING PAPERS

"Can Machine Learning Help Teachers Predict and Improve Student Outcomes? Evidence from a Randomized Controlled Trial" With Jake Meyer.

Can more precise information about students improve teaching and achievement? We randomly provide elementary-school teachers with machine-learning forecasts of students' end-of-year test scores across six charter-school campuses serving 3,068 students in grades 1–6. Linking repeated teacher surveys to administrative monitoring records and test scores, we measure responses in beliefs, student priorities, and instructional practices. Access to forecasts reduces teachers' measured prediction error and increases their reported focus on students predicted to struggle. However, teachers report less independent information gathering and intervention planning, and shift instruction from review toward new material. We estimate an average achievement decline of 0.099 standard deviations, with larger losses among students predicted to score below proficiency. These findings distinguish identifying students likely to struggle from providing effective support: predictive tools should be paired with effective interventions and not evaluated on accuracy alone.

"Association Between Recent Medicaid Expansions and Child Maltreatment Reporting and Outcomes" With Alex Hoagland.

We investigate the association of recent Medicaid expansion in Maine on the volume and quality of child maltreatment reporting. We find that Medicaid expansion led to increased overall reporting, particularly from medical professionals, resulting in increased separations at lower substantiation rates. Associations were stronger among Black and female children. Our results suggest recent Medicaid expansions are associated with increased maltreatment reporting and intervention at lower specificity, resulting in a greater rate of unsubstantiated separations of children from their caregivers.

"Locked Down and Lighting Up: Using COVID-19 Mobility Restrictions as an Instrument for Cigarette Stock and Smoking Intensity" With Don Kenkel and Alan Mathios.

This paper investigates the "temptation effect" – whether increased readily available cigarette stock causally increases smoking intensity. We leverage the unprecedented natural experiment of the COVID-19 pandemic using Nielsen household scanner data, Advan Patterns cell phone data, and USAfacts.org COVID-19 data. An Interrupted Time Series (ITS) model reveals COVID having a sharp and persistent increases in both cigarette stockpiles and smoking rates. Exploration of linked survey data highlights heterogeneous responses related to stress and household characteristics. To isolate the causal effect of stock, we use county-level changes in mobility as an instrumental variable (IV) for changes in cigarette stock. The IV estimates suggest that a one-pack increase in instrumented stock causally increases daily smoking by approximately one cigarette.

WORK IN PROGRESS

"The Impact of Prisoner Education on Recidivism, Labor Market Outcomes, and Public Assistance Utilization" With Valentin Bolotnyy, Mike Lovenheim, and Natalie Millar.

This project evaluates the impact of Career and Technical Education (CTE) programs in Tennessee state prisons on post-release outcomes. We employ an instrumental variables approach leveraging differential exposure to CTE programs based on sentence length and variation in the availability of classes across prisons to instrument for the effect of CTE enrollment.

"Student Group Specific Teacher Learning" With Jason Cook.

This study investigates whether demographic-specific teaching skills are learnable by leveraging a natural experiment in the Columbus City Schools district. In 2003, the district shifted from race-specific to race-blind magnet school lotteries, causing significant changes to the racial composition of its schools. This paper examines how this policy change affected teacher effectiveness for different student populations and across various outcomes. Using a two-way fixed-effect model, the research analyzes how the lottery-induced changes in classroom demographics impacted teacher value-added, calculated separately before and after the policy change. By comparing these effects across different types of student outcomes (e.g., academic, behavioral), the study aims to reveal whether teachers can adapt to demographic shifts and which aspects of teaching effectiveness are most malleable mid-career.

"Improving Teacher Student Matching: Comparative Advantage or Machine Learning"

This paper explores methods to optimize teacher-student pairings by moving beyond single value-added metrics. It investigates which student characteristics (race, sex, prior performance, etc.) offer the greatest potential for gains when matching teachers based on their comparative advantages. The research introduces a novel method for classifying student "types" based on their past performance with different teachers and compares the effectiveness of traditional value-added models against higher-dimensional machine learning predictions for creating optimal classroom assignments. The paper also considers the welfare implications of reassigning students to teachers, rather than teachers to classes, and proposes an RCT to test these matching strategies in a real-world setting.

RESEARCH EXPERIENCE

Cornell University August 2023-Present
Graduate Research Assistant *Ithaca, New York*

- Worked with four professors:
  Mike Lovenheim (Since January 2025)
  Donald Kenkel
  Alan Mathios
  Chen Qiu

Brigham Young University January 2020-July 2021
Pre-doctoral Research Assistant *Provo, Utah*

- Pre-doc with Joe Price
- Collaborated closely on an RCT with Phil Oreopoulos

Brigham Young University April 2019-May 2020
Research Assistant *Provo, Utah*

- Worked with three professors:
  Jeff Denning
  Eric Eide
  Lars Lefgren

TEACHING EXPERIENCE

Cornell Prison Education Program January 2026-May 2026
Sole Instructor *Ithaca, New York*

- Intro to Microeconomics

Brigham Young University Summer 2025
Sole Instructor *Provo, Utah*

- Econ 110: Principles of Economics

Cornell University August 2022-May 2023
Graduate Teaching Assistant *Ithaca, New York*

- Worked with two professors:
  George Boyer: The History of Economic Thought and Institutions
  Stephanie Thomas: Labor Economics

Brigham Young University September 2017-September 2019
Teaching Assistant *Provo, Utah*

- Worked with 6 professors:
  Val Lambson: Game Theory and Economics
  Joe Price: Principles of Economics
  Arden Pope: Principles of Economics
  Jim Kearl: Principles of Economics
  Brennan Platt: Principles of Economics
  Kristina Bishop: Principles of Economics

Brigham Young University September 2017-December 2019
Student Consultant on Teaching *Provo, Utah*

- I worked with professors to improve teaching

SERVICE EXPERIENCE

Cornell University *2024-Present*
Co-Chair: Cornell Heterodox Academy Campus Community Ithaca, New York

SKILLS

Coding Expertise	STATA, Python, Latex
Coding Experience	SQL, MATLAB, R, Git, SLURM, Bash, SASS, Computer Vision
Languages	English (native), Korean (advanced)

ACHIEVEMENTS/AWARDS

Founder's Dissertation Fellowship Award
Sage Fellowship
Deans List (x5)
National Merit Scholar
BYU Racquetball Team Member
Eagle Scout